

Unit 4 Seminar: DR Solutions Design and Review

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Disaster recovery (DR) strategies are typically classified as hot, warm, and cold standby, each reflecting different trade-offs between cost, resilience, and recovery performance. A **hot standby** provides near real-time replication and continuous availability, resulting in near-zero Recovery Time Objective (RTO) and Recovery Point Objective (RPO). However, it is resource-intensive and costly to maintain. A **warm standby** offers a balanced approach with partially active systems and periodic replication, leading to moderate RTO and RPO. In contrast, a **cold standby** relies on backup restoration only after failure, resulting in high RTO and RPO but significantly lower operational cost (Abualkishik et al., 2020).

While these models are conceptually clear, their practical implementation is heavily influenced by underlying technology. Cloud environments enable scalable replication and automated failover, making hot or warm standby architectures more achievable than traditional infrastructure. In contrast, implementing high availability across two on-premise data centres is technically possible but often economically inefficient due to the requirement for synchronous replication, dedicated networking, and continuous infrastructure duplication.

Morrow et al. (2021) argue that cloud adoption introduces vendor lock-in risks, where dependency on proprietary services limits system portability and increases long-term switching costs. This risk can be reduced through multi-cloud strategies, containerisation, and adherence to open standards. However, these approaches may introduce additional architectural complexity. Cloud security risks, including misconfiguration, insecure APIs, and identity mismanagement, remain significant but can be mitigated through strong identity and access management (IAM), encryption, continuous monitoring, and strict application of the shared responsibility model.

Overall, effective DR design requires balancing technical feasibility, cost, and organisational risk appetite rather than relying solely on theoretical recovery classifications.

References

Abualkishik, A. et al. (2020) *Disaster recovery strategies and availability models*. Journal of Cloud Computing.

Morrow, T. et al. (2021) *Cloud computing risks, vendor lock-in and mitigation strategies*. International Journal of Information Systems.

NIST (2020) *Cloud Computing Security Reference Architecture*. National Institute of Standards and Technology.